

Practical 8 CC Lab

```
lab-03-38@lab-03-38-OptiPlex-3280-AIO:~$ vim postfix275.l
lab-03-38@lab-03-38-OptiPlex-3280-AIO:~$ vim postfix275.y
lab-03-38@lab-03-38-OptiPlex-3280-AIO:~$ yacc -d postfix275.y
lab-03-38@lab-03-38-OptiPlex-3280-AIO:~$ lex postfix275.l
lab-03-38@lab-03-38-OptiPlex-3280-AIO:~$ gcc y.tab.c lex.yy.c -o postfix275 -ll
lab-03-38@lab-03-38-OptiPlex-3280-AIO:~$ ./postfix275
Postfix Evaluator: Enter expressions (Ctrl+C to exit)
3 4 +
Result = 7
10 20 * 5 /
Result = 40
10 0 /
Error: Division by zero
Result = 0
10+5
Syntax Error! Please re-enter.
12 50 * 95 + 10 2 / -
Result = 690
^C
lab-03-38@lab-03-38-OptiPlex-3280-AIO:~$ vim postfix275.l
lab-03-38@lab-03-38-OptiPlex-3280-AIO:~$ vim postfix275.y
lab-03-38@lab-03-38-OptiPlex-3280-AIO:~$
```

```
%{
#include "y.tab.h"
#include <stdlib.h>
}%

%%
[0-9]+      { yylval = atoi(yytext); return NUMBER; }
[ \t]       { /* Ignore spaces and tabs */ }
\n          { return '\n'; }
.           { return yytext[0]; }
%%

int yywrap() {
    return 1;
}

~
~
~
~
```

Practical 8 CC Lab

```
%{
#include <stdio.h>
#include <stdlib.h>
int yylex(void);
void yyerror(char *s);
%}

%token NUMBER

%%

lines: /* empty */
| lines line
;

line: '\n'
| exp '\n' { printf("Result = %d\n", $1); }
| error '\n' { yyerror("Syntax Error! Please re-enter.\n"); }
;

exp: exp exp '+' { $$ = $1 + $2; }
| exp exp '-' { $$ = $1 - $2; }
| exp exp '*' { $$ = $1 * $2; }
| exp exp '/' {
    if($2 == 0) {
        printf("Error: Division by zero\n");
        $$ = 0;
    } else {
        $$ = $1 / $2;
    }
}
| NUMBER { $$ = $1; }
;

void yyerror(char *s) {
    /* yyerror is handled manually in the rules for better recovery */
}

int main() {
    printf("Postfix Evaluator: Enter expressions (Ctrl+C to exit)\n");
    yyparse();
    return 0;
}
```